

WANTED - Wallet-friendly electronic Registered Delivery³

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Abstract: The “European Business Wallet” (EBW) is intended to complement the “European Digital Identity Wallet” (EUDIW) and must support Qualified Electronic Registered Delivery Services (QERDS). Current QERDS standards under eIDAS and ETSI EN 319 52x predate wallet-based infrastructures, rely on XML or S/MIME, and do not fully exploit the cryptographic capabilities of wallets. The present paper reviews these standards in the context of the EUDIW and EBW and identifies needs for revision to support future wallet-based services and solutions.

Keywords: European Business Wallet (EBW), eIDAS, European Digital Identity Wallet (EUDIW) (Qualified) Electronic Registered Delivery Service ((Q)ERDS), Verifiable Credential (VC), Wallet

1 Introduction

On 29 January 2025, the European Commission presented the “Competitiveness Compass” as a roadmap to strengthen Europe’s competitiveness and support the development of future technologies, services, and products [EC25a; EC25b]. Among the flagship actions announced for 2025 was the European Business Wallet (EBW), envisioned as a key enabler for simple and digital business interactions with public administrations across the EU. Following a public consultation [EBWC25], the Commission published on 19 November 2025 its proposal for a Regulation establishing European Business Wallets [EBWR25].

As explained in Section 2, the EBW is intended to support a broad range of trust services, including authentication, owner identification using “European Business Wallet owner identification data” (EBWoid), electronic attestations of attributes, signatures, seals, time stamps, mandates, and Qualified Electronic Registered Delivery Services (QERDS) under Article 3 (37) eIDAS for trustworthy and legally binding data exchange.

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³ Note, that this document is an *extended abstract* only to meet the given page limits. Please reach out to the lead author to request the full version of the paper.

The remainder of this paper is structured as follows. Section 2 recalls the regulatory and technical background of the EBW and QERDS. Section 3 examines whether and how existing QERDS-related standards can be used with the EBW and the EUDIW. Section 4 concludes with the main findings and an outlook on future developments.

2 Background

2.1 Background on Qualified Electronic Registered Delivery Services

2.1.1 Requirements specified in Article 44 of (EU) No. 910/2014

According to Article 3 (36) of the eIDAS Regulation (EU) No. 910/2014, an Electronic Registered Delivery Service (ERDS) enables the electronic transmission of data between third parties while providing evidence of handling, including proof of sending and receiving, and protecting the data against loss, theft, damage, or unauthorised alteration. A Qualified Electronic Registered Delivery Service (QERDS), as defined in Article 3 (37), is an ERDS that complies with the requirements of Article 44.

Under Article 44 (1), a QERDS must be provided by one or more qualified trust service providers, ensure reliable identification of sender and addressee, secure transmission by means of advanced electronic signatures or seals, indicate any changes to the data, and provide qualified electronic time stamps for sending, receiving, and modifications. Where data is transferred between multiple qualified trust service providers, these requirements apply to all of them.

2.1.2 Requirements specified in CIR (EU) 2025/1944 with respect to identification

According to Article 44 (1a) of the eIDAS-Regulation, compliance with the QERDS requirements is presumed if the service conforms to the standards, specifications, and procedures referenced in the corresponding implementing act under Article 44(2), namely Commission Implementing Regulation (CIR) (EU) 2025/1944.

Article 1 and Annex I of this regulation require compliance with the policy and security requirements of ETSI EN 319 521, while introducing *notable adaptations* to the identification requirements of Article 44 (1) (b) and (c) of the eIDAS-Regulation, which are displayed in the full version of the paper.

Identification of the Sender. A key observation is that Article 44 (1) (b) requires identification of the sender with a “high” level of confidence, whereas CIR (EU) 2025/1944 translates this into a “Level of Assurance” (LoA) of only “substantial” in cases of prior

physical presence, or the use of an advanced electronic signature or seal based on a certificate with a “Normalised Certificate Policy” (NCP) according to [ETSI EN 319 411-1](#).

Identification of the Recipient. For the recipient, the regulation is stricter. While Article 44 (1) (c) merely requires identification before delivery, CIR [\(EU\) 2025/1944](#) demands verification with a “very high” level of confidence. In practice, this requires LoA “high”, the use of the EUDIW, or a qualified electronic signature or seal.

2.1.3 Other Requirements specified in ETSI EN 319 521 and CIR (EU) 2025/1944

In addition to the requirements for the identification of the sender and recipient summarised above, the implementing act CIR [\(EU\) 2025/1944](#) adapts a few additional policy and security requirements from [ETSI EN 319 521](#), which profiles the general policy and security requirements from [ETSI EN 319 401 \(v3.1.1\)](#)⁴.

2.1.4 Electronic Registered Delivery Services Reference Architecture

The “Electronic Registered Delivery Services Reference Architecture” defined in [ETSI EN 319 522-1](#) is best recalled based on Figure 4 and Figure 7 in this standard.

The system consists of a *Sender*, which uses its *S-ERDS*, and the *Recipient*, which uses its *R-ERDS*. The main data exchange happens over the *ERDS-RI* interface between the two ERDS and the commonly used *Shared Infrastructure* allows to find the necessary communication endpoints and provides additional trust-related information.

It is important to understand, that in the current ETSI EN 319 52x series of standards only the *ERDS-RI* interface is specified in a sufficient level of detail such that one can hope for interoperability at this point. Furthermore, semantic content specifications outlined in [ETSI EN 319 522-2](#) are currently only mapped to metadata formats in [ETSI EN 319 522-3](#) for

- *XML-based* data exchange according to [OASIS AS4](#) together with the corresponding bindings in [ETSI EN 319 522-4-1](#), [ETSI EN 319 522-4-2](#), [ETSI EN 319 522-4-3](#) and to metadata formats for
- *email-based* data exchange according to [RFC 5322](#) and [ETSI EN 319 522-3](#).

More modern metadata formats and bindings for JSON-based data exchange for example, with contemporary identity management standards, such as [OpenID Connect](#) or even

⁴ Note, that CIR [\(EU\) 2025/1944](#) points to the specific version 3.1.1, while there is a more recent version 3.2.1 of the standard [ETSI EN 319 401 \(v3.2.1\)](#) and the next update to version 3.3.1 is planned to be produced within the scope of the recently started [STF 705 on “EUDI Wallet Standardisation and Technical Specifications”](#).

[OpenID4VCI](#) and [OpenID4VP](#), are unfortunately not yet existing and hence need to be subject of future standardization work.

2.2 Background on European Business Wallets

According to Article 7 of [\[EBWR25\]](#), European Business Wallets (EBWs) are provided by notified “Providers of European Business Wallets” (PoEBW). These providers must notify themselves to the national competent supervisory authority and are then included in an official list. PoEBWs must at least comply with Article 19a of the eIDAS-Regulation for non-qualified trust service providers or, if qualified, fulfill Article 24 of the eIDAS-Regulation. In all cases, they must also comply with the NIS2 Directive [\(EU\) 2022/2555](#).

Under Article 8 of [\[EBWR25\]](#), each EBW is equipped with “European Business Wallet owner identification data” (EBWoid), a specific electronic attestation of attributes⁵ containing at least the official name of the economic operator or public sector body and its European Unique Identifier (EUID). Since the proposed regulation requires EBWs to issue electronic attestations of attributes, they will likely need suitable X.509 certificates in line with [RFC 5280](#), [ETSI TS 119 472-1](#), [ETSI TS 119 472-3](#). If an EBW also acts as a Wallet-Relying Party, the EBWoid could be or correspond to the Wallet-Relying Party Access Certificate defined in CIR [\(EU\) 2025/848](#) and [ETSI TS 119 411-8](#).

Article 5(1) of [\[EBWR25\]](#) defines the core EBW functionalities. These include issuing, storing, sharing, and presenting electronic attestations of attributes; selective disclosure of EBWoid and attributes; secure exchange of data with other EBWs, EUDIWs, and relying parties; support for qualified signatures, seals, and time stamps; issuance of linked attestations; authentication; QERDS-based transmission and reception of documents and data; management of user and relying-party authorisations; data export; transaction logging; and access to a common dashboard for QERDS communications.

Article 7 of [\[EBWR25\]](#) foresees a “European Digital Directory” (EDD), which is expected to be established and maintained by the European Commission as a trusted source of EBW-related metadata accessible via API and secure web interface.

The most important aspects are visualized in Fig. 1 above and are briefly summarized in the following:

- The EBW is equipped with an EBWoid, which may be in form of a WRPAC according to [ETSI TS 119 411-8](#) for example.
- This allows to request verifiable presentations from an EUDIW or another EBW via [OpenID4VP](#) and issue verifiable credentials to an EUDIW or another EBW via [OpenID4VCI](#).

⁵ Note, that an X.509 certificate is also an electronic attestation of attributes in the sense of Article 3 (44) of the eIDAS-Regulation.

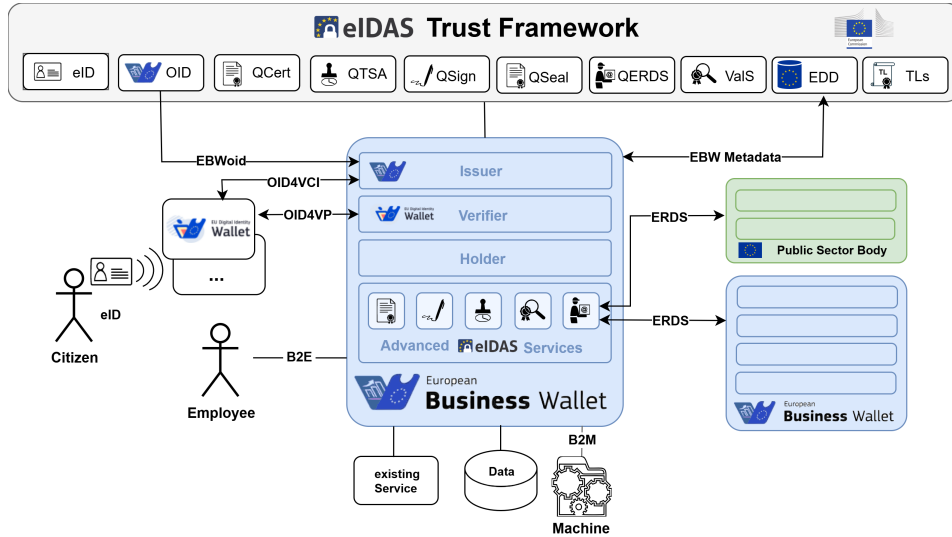


Fig. 1: System Architecture with European Business Wallet according to [EBWR25]

- The EBW is able to create and validate⁶ advanced and qualified electronic signatures and seals and request (qualified) time stamps.
- The EBW is able to transmit and receive electronic documents and data in a way, which supports confidentiality and integrity, whereas the communication endpoints of EBWs can be looked up in the EDD.

While the latter could take place by using a QERDS, as suggested by [EBWR25, Article 5 (1) (i)], it is rather obvious that the EBW is equipped with the necessary functionality for trustworthy data exchange itself, *without* the detour to invoke an external QERDS.

3 Analysis

The main observations of this paper can be summarized as follows:

- Rather than relying on an external QERDS, as suggested in [EBWR25, Article 5 (1) (i)], an EBW appears to be very well suited to support trustworthy data exchange through its own core functions, using encryption, electronic seals and qualified time stamps.
- The *Shared Infrastructure* referred to in ETSI EN 319 521 can easily be realised in a standardised way using the existing Trusted Lists according to Article 22 of the

⁶ Note, that the validation of advanced and qualified electronic seals is not explicitly mentioned in [EBWR25], but it is clear, that this kind of functionality is needed for conducting trustworthy electronic business processes.

eIDAS-Regulation and [ETSI TS 119 612](#), the new Lists of Trusted Entities (LoTE) under CIR ([EU](#)) [2024/2980](#) ETSI TS 119 602, and especially the planned EDD [[EBWR25](#)], which is expected to provide a search API.

- Although the *ERDS-RI* could in principle be used as currently specified in the present ETSI EN 319 52x series of standards, it would be strongly advisable to extend the current set of standards with an additional JSON-based binding that is more suitable for wallet-based architectures and also look into alternative and more powerful communication protocols, such as the “Eclipse Dataspace Protocol” (DSP) according to [DSP 2025-01](#) and [ISO/IEC DIS 26450](#) with eIDAS-based credentials as proposed within [[WHH+25](#)] in more detail.
- In addition to the obvious EBW-to-EBW use case, it is also recommended to consider the EBW-to-EUDIW use case as well, as this communication pattern is also necessary in equally important B2C and G2C scenarios.

4 Conclusion and Outlook

The present paper has analysed the legal requirements and existing standards for QERDS in the context of the EBW and concludes that the main functions of a QERDS can largely be provided by the EBW itself, with additional functionality mainly limited to obtaining suitable time stamps.

It also identified gaps in the current standardization landscape and outlined possible approaches to address them, which may be taken forward in relevant standardization bodies such as [ETSI ESI](#) for example.

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